

Merging Dynamic Covariates: Predicting density and assessing factors driving marbled murrelet distribution in Puget Sound using hierarchical distance sampling

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Required Packages

```
library(ggplot2) # plotting
library(sf) # spatial
library(tidyverse)
```

Required Files Previously Made or Downloaded

- grid2km_hex.gpkg
- lat_long_times.rds
- grid_tides.rds (made in “04-TidalData”)
- track_tides.rds (made in “04-TidalData”)
- Dynamicgrid_data_NEW.RData (made in “03-DynamicCovariates”)

```
hex_grids <- st_read("GIS_COVS/grid2km_hex.gpkg")

tracklines <- st_read("../GIS_COVS/all_tracklines_2026_final.gpkg")
lat longs <- readRDS("../GIS_COVS/lat_long_times.rds")

tidal_grids <- readRDS("GIS_COVS/grid_tides.rds")

tidal_tracks <- readRDS("GIS_COVS/track_tides.rds")

load("GIS_COVS/Dynamicgrid_data_NEW.RData")
```

Tidal

```
monyears <- unique(lat longs$monyear)
length(monyears)
```

```
## [1] 100
```

```
tidal_gclean <- tidal_grids %>%
  mutate(monyear = toupper(monyear)) %>%
  filter(monyear %in% monyears)

length(unique(tidal_gclean$monyear))
```

```
## [1] 100
```

```
length(unique(tidal_gclean$ID))
```

```
## [1] 1545
```

```
tidal_tclean <- tidal_tracks %>%  
  mutate(moneyear = toupper(moneyear)) %>%  
  filter(moneyear %in% monyears)  
  
length(unique(tidal_tclean$moneyear))
```

```
## [1] 100
```

```
length(unique(tidal_tclean$trip_date))
```

```
## [1] 8095
```

Grid Chem

```
chem_clean <- hex_chem %>%  
  st_drop_geometry() %>%  
  dplyr::select(-area) %>%  
  pivot_longer(  
    cols = matches("^do|dic."),  
    names_to = c("source", "mon", "year"),  
    names_pattern = "(do|dic).([A-Z]{3}).(\\d{4})" %>%  
    mutate(moneyear = paste0(mon, "-", year)) %>%  
    filter(source == "do" & moneyear %in% monyears) %>%  
    dplyr::select(-source) %>%  
    rename(do = value)  
  
length(unique(chem_clean$moneyear))
```

```
## [1] 100
```

```
length(unique(chem_clean$ID))
```

```
## [1] 1545
```

Track Chem

```
chem_clean_track <- track_chem %>%  
  st_drop_geometry() %>%  
  dplyr::select(-total_length) %>% # Remove poly_overlapkm2  
  pivot_longer(  
    cols = matches("^do|dic."),  
    names_to = c("source", "mon", "year"),
```

```

names_pattern = "(do|dic).([A-Z]{3}).(\\d{4})" %>%
mutate(moneyear = paste0(mon,"-",year),
      moneyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
filter(source == "do" & moneyear == moneyear2) %>%
dplyr::select(-c(source, moneyear2)) %>%
rename(do = value) %>%
group_by(trip_date, date, mon, year, moneyear) %>%
summarise(do = mean(do), .groups="drop") %>%
ungroup()

length(unique(chem_clean_track$moneyear))

```

```
## [1] 100
```

```
length(unique(chem_clean_track$trip_date))
```

```
## [1] 8095
```

Grid Phy

```

phy_clean <- hex_phy %>%
  st_drop_geometry() %>%
  dplyr::select(-area) %>% # Remove poly_overlapkm2
  pivot_longer(
    cols = matches("(ss|st|sig)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "(ss|st|sig).([A-Z]{3}).(\\d{4})" %>%
  mutate(moneyear = paste0(mon,"-",year)) %>%
  filter(moneyear %in% moneyears) %>%
  pivot_wider(names_from = source,
    values_from = value)

length(unique(phy_clean$moneyear))

```

```
## [1] 100
```

```
length(unique(phy_clean$ID))
```

```
## [1] 1545
```

Track Phy

```

phy_clean_track <- track_phy %>%
  st_drop_geometry() %>%
  dplyr::select(-total_length) %>% # Remove poly_overlapkm2
  pivot_longer(
    cols = matches("(ss|st|sig)."),

```

```

    names_to = c("source", "mon", "year"),
    names_pattern = "^(ss|st|sig).([A-Z]{3}).(\\d{4})" %>%
mutate(moneyear = paste0(mon,"-",year),
      moneyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
filter(moneyear == moneyear2) %>%
pivot_wider(names_from = source,
  values_from = value) %>%
select(-moneyear2) %>%
group_by(trip_date, date, mon, year, moneyear) %>%
summarise(st = mean(st), ss = mean(ss), .groups="drop") %>%
ungroup()

length(unique(phy_clean_track$moneyear))

```

```
## [1] 100
```

```
length(unique(phy_clean_track$trip_date))
```

```
## [1] 8095
```

```

bio_clean <- hex_bio %>%
  st_drop_geometry() %>%
  dplyr::select(-area) %>%
  pivot_longer(
    cols = matches("(z1|z2|dia)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "^(z1|z2|dia).([A-Z]{3}).(\\d{4})" %>%
mutate(moneyear = paste0(mon,"-",year)) %>%
filter(moneyear %in% moneyears) %>%
pivot_wider(names_from = source,
  values_from = value)

length(unique(bio_clean$moneyear))

```

Grid Bio

```
## [1] 100
```

```
length(unique(bio_clean$ID))
```

```
## [1] 1545
```

```

bio_clean_track <- track_bio %>%
  st_drop_geometry() %>%

```

```
dplyr::select(-total_length) %>%
pivot_longer(
  cols = matches("^(z1|z2|dia)."),
  names_to = c("source", "mon", "year"),
  names_pattern = "^(z1|z2|dia).([A-Z]{3}).(\\d{4})" %>%
  mutate(moneyear = paste0(mon,"-",year),
         moneyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
filter(moneyear == moneyear2) %>%
pivot_wider(names_from = source,
            values_from = value) %>%
select(-moneyear2) %>%
group_by(trip_date, date, mon, year, moneyear) %>%
summarise(dia = mean(dia), z1 = mean(z1), z2 = mean(z2), .groups="drop") %>%
ungroup()
```

```
length(unique(bio_clean_track$moneyear))
```

Track Bio

```
## [1] 100
```

```
length(unique(bio_clean_track$trip_date))
```

```
## [1] 8095
```

```
turb_clean <- hex_par %>%
  st_drop_geometry() %>%
  dplyr::select(-area) %>%
  pivot_longer(
    cols = matches("^(turb|par)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "^(turb|par).([A-Z]{3}).(\\d{4})" %>%
  mutate(moneyear = paste0(mon,"-",year)) %>%
  filter(moneyear %in% moneyears) %>%
  pivot_wider(names_from = source,
            values_from = value)
```

```
length(unique(turb_clean$moneyear))
```

Grid PAR

```
## [1] 100
```

```
length(unique(turb_clean$ID))
```

```
## [1] 1545
```

```

turb_clean_track <- track_par %>%
  st_drop_geometry() %>%
  dplyr::select(-total_length) %>%
  pivot_longer(
    cols = matches("^ (turb|par)."),
    names_to = c("source", "mon", "year"),
    names_pattern = "^ (turb|par).([A-Z]{3}).(\\d{4})" %>%
    mutate(monyear = paste0(mon, "-", year),
           monyear2 = toupper(format(as.Date(date), "%b-%Y"))) %>%
  filter(monyear == monyear2) %>%
  pivot_wider(names_from = source,
             values_from = value) %>%
  select(-monyear2) %>%
  group_by(trip_date, date, mon, year, monyear) %>%
  summarise(par = mean(par), .groups="drop") %>%
  ungroup()

length(unique(turb_clean_track$monyear))

```

Track PAR

```
## [1] 100
```

```
length(unique(turb_clean_track$trip_date))
```

```
## [1] 8095
```

Combining all data

```

dynamic_data_grid <- bio_clean %>%
  full_join(turb_clean, by = c("ID", "numid", "monyear", "year", "mon")) %>%
  full_join(phy_clean, by = c("ID", "numid", "monyear", "year", "mon")) %>%
  full_join(chem_clean, by = c("ID", "numid", "monyear", "year", "mon")) %>%
  full_join(tidal_gclean, by = c("ID", "monyear")) %>%
  full_join(hex_suit %>%
    dplyr::select(-c(layer, area)) %>%
    st_drop_geometry(),
    by = c("ID", "numid", "year"))

dynamic_data_track <- bio_clean_track %>%
  full_join(turb_clean_track, by = c("trip_date", "date",
                                     "monyear", "year", "mon")) %>%
  full_join(phy_clean_track, by = c("trip_date", "date",
                                     "monyear", "year", "mon")) %>%
  full_join(chem_clean_track, by = c("trip_date", "date",
                                     "monyear", "year", "mon")) %>%
  full_join(tidal_tclean, by = c("trip_date", "monyear")) %>%

```

```
full_join(track_suit %>%  
  dplyr::select(-c(total_area)) %>%  
  st_drop_geometry(),  
  by = c("trip_date", "date", "year"))
```

Save

```
saveRDS(dynamic_data_grid, "dynamic_data_clean_grids_NEW.rds")  
saveRDS(dynamic_data_track, "dynamic_data_clean_tracks_NEW.rds")
```